Tutorial 3 - I/O Ports and Device Interfacing (Part 1)

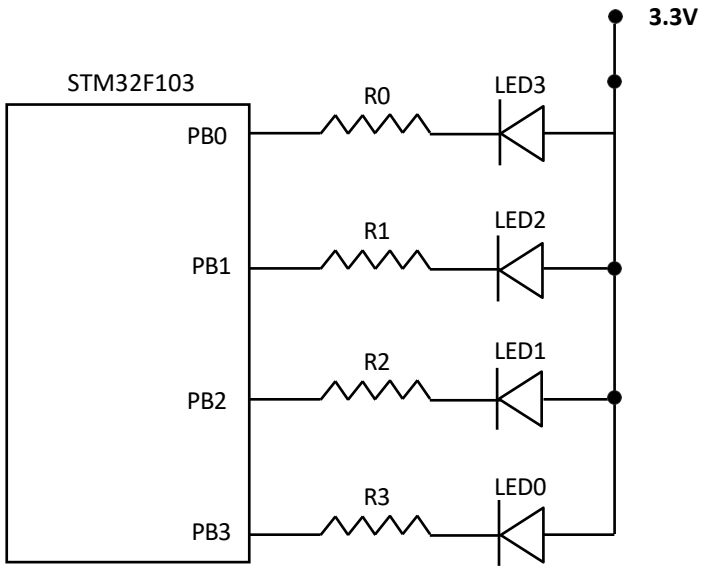
Q3-1.1 What is the expected result after the line of C code below has been executed?

```
PortOut PTB(PortB, 0x0000000F);
```

(a) Port B pins 15 to 4 are configured as outputs, while Port B pins 0 to 3 are configured as inputs.
(b) Port B pins 15 to 4 are ignored, while Port B pins 0 to 3 are configured as inputs.
(c) Port B pins 15 to 4 are ignored, while Port B pins 0 to 3 are configured as outputs.
(d) Port B pins 15 to 4 are configured as inputs, while Port B pins 0 to 3 are configured as outputs.

C

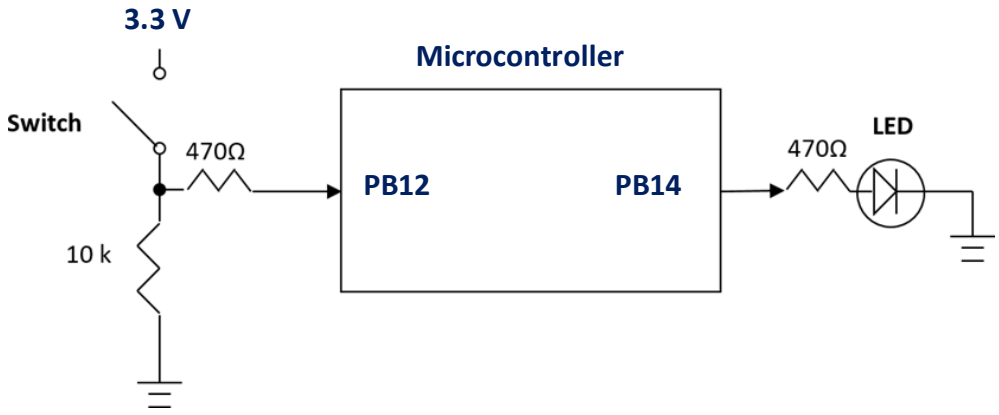
Q3-1.2 Based on the circuit diagram below, what is the correct value to write to PORTB to switch on LED1 ONLY?



(a) 0x04
(b) 0x01
(c) 0x08
(d) 0x0B

d

Q3-1.3 Based on the circuit diagram below, answer the following questions.



(a) Is the switch connected as an Active High or Active Low input to the microcontroller?

(b) When the switch is closed, what is the voltage level at microcontroller pin PB12?

(c) Assuming that the LED has a voltage drop of 1.8 volts when turned ON, what is the current supplied by the microcontroller at output pin PB14?

(d) Implement C code to fulfil the following requirements,

1. when the switch is closed, the LED shall blink at a frequency of 1 Hz.
2. when the switch is opened, the LED shall be turned OFF.

a) Active High

b) 3.3V

c)
I = V/R
I = (3.3-1.8)V/470 ohms
I = 3.19mA

d)

```
#include "mbed.h"

DigitalIn Switch(PB_12)
DigitalOut LED(PB14)

int main(){
while (true) {
    if (Switch.read()== 1){
    led = !led
    thread_sleep_for(500)
} else {
   led = 0
 }
}
```